

Week 1 Lab — Python Foundations

From the Big Picture to Your First Analysis · Total: 35 marks (+3 bonus) · Due: end of Week 1

Name: _____ Student ID: _____ Program: _____

Instructions

- Work in a Jupyter or Google Colab notebook named **Week1_YourName**. Answer every exercise in a separate code cell, in order.
- The dataset for Part 5 is provided in **week1_students.csv** and as ready-to-run Python code in the lab notebook.
- **Submit before the end of the week:** (1) your completed notebook, (2) screenshots of your running code, (3) a ½–1 page report describing what you built and one thing that surprised you.
- You may discuss with classmates, but every line of code you submit must be typed and understood by **you**.

Part 0 · Warm-Up (not graded)

Create a new notebook, rename it **Week1_YourName**, and run:

```
print("Hello, Big Data!")
```

Part 1 · Variables & Data Types (5 marks)

Exercise 1.1 (2 marks)

Create four variables about **yourself**: `my_name` (str), `my_age` (int), `my_gpa` (float — invent one), and `am_i_present` (bool). Print each variable together with its type.

Hint: `print(my_name, type(my_name))`

Exercise 1.2 — Type conversion (2 marks)

Real datasets often store numbers as text. Starting from the given line below, convert the value to an `int`, add 1 (a birthday!), and print the result.

```
age_text = "21"      # given: a number trapped inside text
```

Question 1.3 — Think about it (1 mark)

What does `"3" + "4"` produce? Run it, then explain in one sentence why the answer is not 7.

Part 2 · Strings — Working with Text (5 marks)

Exercise 2.1 — f-strings (2 marks)

Using your variables from Exercise 1.1, print the following sentence with **one f-string**:

I am NAME, AGE years old, studying at AUCA.

Reminder: `f"Hello {my_name}"` — variables go inside `{ }`

Exercise 2.2 — String tools (3 marks)

Given `course = "Big Data Analytics"`, print: (a) the course name in UPPERCASE, (b) the number of characters it contains, (c) only the first 3 characters.

Hints: `.upper()` · `len(...)` · slicing `[0:3]`

Part 3 · Lists — Your First Descriptive Statistics (7 marks)

Exercise 3.1 (4 marks)

The quiz scores of your 5-person study group are given below. Print: (a) the first and the last score, (b) the number of scores, (c) the average, (d) the highest and the lowest score.

```
scores = [72, 85, 91, 64, 78]
```

Hints: `scores[0]` · `scores[-1]` · `len()` · `sum()/len()` · `max()` · `min()`

Exercise 3.2 — Lists change (3 marks)

A new member joins with a score of **88**, and you discover the 4th score (position 3) should be **66**, not 64. (a) Append 88 to the list, (b) replace position 3 with 66, (c) print the updated list and its new average.

Part 4 · Dictionaries — Labeled Data (5 marks)**Exercise 4.1 (2 marks)**

Build a dictionary called `me` describing yourself with the keys "name", "age", "program", "district". Print your program using `me["program"]`.

Exercise 4.2 — Update and extend (3 marks)

(a) Add a new key "score" with a value of your choice, (b) increase "age" by 1, (c) print all the keys with `me.keys()` and then the whole dictionary.

Part 5 · Class Dataset Analysis (13 marks)

You will analyze the dataset below — 20 AUCA students. It is provided in **week1_students.csv** (open it in Excel to see the table) and as ready-to-run Python code in the lab notebook: a **list of 20 dictionaries** called `students`.

ID	Name	Age	Gender	Program	District	Att. %	Score
AUCA001	Aline Uwase	21	F	Information Systems	Gasabo	92	85
AUCA002	Eric Niyonzima	23	M	Software Engineering	Kicukiro	88	78
AUCA003	Diane Mukamana	20	F	Information Systems	Nyarugenge	95	91
AUCA004	Jean Bosco Habimana	24	M	Networking	Gasabo	70	64
AUCA005	Clarisse Ingabire	22	F	Accounting	Musanze	85	72
AUCA006	Patrick Mugisha	21	M	Software Engineering	Huye	90	83
AUCA007	Sandrine Umutohi	19	F	Information Systems	Kicukiro	97	94
AUCA008	Emmanuel Nsengiyumva	25	M	Networking	Rubavu	65	58
AUCA009	Grace Uwimana	20	F	Accounting	Gasabo	82	76
AUCA010	Yves Ndayisenga	22	M	Software Engineering	Nyarugenge	78	69
AUCA011	Josiane Mukeshimana	21	F	Information Systems	Muhanga	93	88
AUCA012	Claude Iradukunda	23	M	Networking	Gasabo	75	66
AUCA013	Beata Nyirahabimana	24	F	Accounting	Nyamagabe	89	81
AUCA014	Olivier Twagirimana	20	M	Software Engineering	Kicukiro	96	90
AUCA015	Chantal Uwera	22	F	Information Systems	Rusizi	68	61
AUCA016	Fabrice Byiringiro	21	M	Networking	Musanze	84	74
AUCA017	Solange Ishimwe	19	F	Software Engineering	Gasabo	91	87
AUCA018	David Hakizimana	25	M	Accounting	Nyagatare	72	55
AUCA019	Immaculee Uwamahoro	23	F	Information Systems	Kicukiro	87	79
AUCA020	Samuel Rukundo	20	M	Networking	Nyarugenge	94	92

The notebook also gives you three **ready-made column lists** (run the provided cell — the technique behind it is explained in Week 5):

```
names      = [s["name"] for s in students]      # every student's name
all_scores = [s["score"] for s in students]      # every score
districts  = [s["district"] for s in students]  # every district
```

Exercise 5.1 — Describe the class (4 marks)

Using f-strings, print: (a) how many students are in the dataset, (b) the class average score rounded to 1 decimal — `round(..., 1)`, (c) the highest and lowest score, (d) the range (highest minus lowest).

Exercise 5.2 — Inspect individual records (3 marks)

(a) Print the full record of the **first** and the **last** student. (b) For the student at position 6, print one f-string sentence: *NAME from DISTRICT studies PROGRAM and scored SCORE*.

Exercise 5.3 — Count by category (3 marks)

Lists have a built-in counter: `districts.count("Gasabo")`. Print how many students come from (a) Gasabo, (b) Kicukiro, and (c) one other district of your choice.

Exercise 5.4 — Stretch goal: who is the top student? (3 marks + 3 bonus)

Without any loop, print: *The top student is NAME with a score of SCORE*. Combine three things you already know:

1. `max(all_scores)` → the best score 2. `all_scores.index(...)` → its position 3. `names[position]` → the name

Bonus (+3): do the same for the **LOWEST** score, and print both students' attendance too.

Part 6 · Reflection (required, checked for completion)

1. Is this 20-student dataset **Big Data**? Which of the 5 V's does it clearly *not* have yet? Explain briefly.
2. Describe one thing from today's session that surprised or confused you.

✓ Submission checklist

- ☐ Notebook renamed to Week1_YourName
- ☐ All exercises run without errors
- ☐ Reflection answered
- ☐ Screenshots of running code
- ☐ ½–1 page report
- ☐ Submitted before the end of Week 1

Marking scheme: Part 1 — 5 · Part 2 — 5 · Part 3 — 7 · Part 4 — 5 · Part 5 — 13 · Total 35 (+3 bonus). Reflection and checklist are required for the submission to be graded.